

David Dangerfield

Principal Software Engineer | Elite Velocity & Quality via AI + Experience

Consumer Products | Payments | Healthcare | Medical Devices | 20+ Years

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SUMMARY

High-output Principal Software Engineer with over 20 years of experience delivering robust features, eliminating technical debt, and modernizing complex web/app experiences supporting multi-billion-dollar revenue streams. Expert individual contributor leveraging advanced agentic AI to achieve elite delivery velocity while maintaining critical human oversight for compliance, security, and correctness. Proven track record of executing rapid cross-repository updates and library migrations under regulatory (FDA, HIPAA) constraints. Strong computer science foundation (MS, Purdue) and real-world expertise spanning backend, frontend, test architecture, and systems engineering.

Technical Profile

Languages: TypeScript, JavaScript, Java, Kotlin, Python

Backend & Frameworks: REST APIs, Spring Boot, Node.js, Message Queues, HIPAA-Compliance, Microservices, I18n, Legacy Modernization

Data Stores: PostgreSQL, MySQL, MongoDB, SQL Server, Redis

Cloud & Infrastructure: AWS, Azure DevOps, Google Cloud Platform, GitHub Actions, Docker, Kubernetes

Testing, Tools, & AI: Copilot, Claude, Playwright, Cypress, Jest, JUnit

Frontend: Vue.js, React, Next.js, GraphQL, Angular, Design Systems, Accessibility, Component Libraries, Tailwind, eCharts

EXPERIENCE

Principal Software Engineer (Contractor)

Mar 2025 - Present

[Patterson Companies, Inc. \(Remote\)](#)

- Implemented a rewrite of the rigid legacy Angular.js bill pay frontend for dental and veterinary sales sites to responsive Vue.js, supporting a multi-billion dollar revenue stream with locale-aware code
- Translated Figma designs to TypeScript and Vue.js frontend extending the UI component library for consistency and integrating with .NET backend hosted in Azure and C# microservices to SAP
- Transitioned to agentic development using GitHub Copilot and Claude

Tech: Copilot, Claude, Node.js, TypeScript, Vue.js, Microfrontends, Microservices, SQL Server, Azure DevOps, I18n

Lead Software Developer (Contractor)

Jul 2022 - Mar 2025

[Life Time Inc. \(Remote\)](#)

- Built web and app experiences for a major consumer fitness platform, driving hundreds of thousands of new accounts and app installs
- Replaced legacy libraries in core platform code for high-traffic consumer sites

Tech: GitHub Copilot, TypeScript, Vue.js, Bootstrap, Vite, Vitest, Playwright, Legacy Modernization, Mobile App

Senior Software Engineer (Concurrent Part-Time Contractor)

Feb 2018 - Mar 2024

[Nurocor - Digitalized Clinical Development \(Remote\)](#)

- Developed UI for a structured authoring platform for FDA clinical trial documentation, enabling compliant document templates, formal review workflows, and complete audit trails on every edit
- Implemented audit trail UI functionality ensuring full traceability, a core FDA requirement
- System recognized by the TransCelerate BioPharma ClinEco Protocol Review Challenge as Best Integrated Protocol Authoring and Review Solution

Tech: Kubernetes, HIPAA, Vue.js, Nuxt.js, Storybook, GraphQL, Component Libraries

Principal Software Engineer Consultant (Contractor)

Jan 2021 - Jul 2022

[NextEra Analytics \(Remote\)](#)

- Modernized outdated processes by implementing web application workflows to replace manual spreadsheet work
- Developed interactive executive dashboards and complex data visualizations using eCharts to make capital allocation data easily actionable

Tech: TypeScript, JavaScript, Python, Terraform, AWS, Angular, Vue.js, GraphQL, eCharts, Storybook

Senior Software Consultant (Concurrent Part-Time Contractor) *Sep 2020 - Feb 2022*

Intricately (Acq. by HG Insights) (Remote)

- Engaged directly by the CTO as an expert frontend advisor to audit legacy code, deliver actionable technical recommendations, and elevate engineering practices across an international team.
- Provided implementation guidance to ensure a smooth, future-proof Vue 2 to Vue 3 migration path while actively reducing codebase complexity and preventing over-engineering.
- Collaborated closely with remote developers to build trust, mentor, and provide leadership.

Tech: TypeScript, Vue.js

Senior Software Engineer

Sep 2018 - Jan 2021

SmugMug (Remote)

- Developed and maintained SmugMug.com, one of the longest-running independent subscription-based web services, hosting billions of photos for millions of customers worldwide
- Executed the migration of legacy frontend payment code to Stripe to modernize transaction processing for a high-volume website

Tech: AWS, Node, JavaScript, React, Tailwind, GraphQL, Continuous Deployment, Legacy Modernization, Accessibility

Advisory Software Engineer, Watson Health, & IBM Services

Nov 2015 - Aug 2018

IBM Services Group (Rochester, MN)

- Developed early AI-powered pharmaceutical research experiences using IBM Watson NLP to surface relationships across biomedical literature
- Built clinical decision support applications for healthcare workers including diabetes treatment recommendations and patient history visualization, under HIPAA compliance protocols
- Delivered cognitive AI applications for financial clients including a top 50 US bank
- Co-founded FED@IBM Rochester, an internal frontend developer community of 30+ engineers hosting twice-monthly sessions on JavaScript, accessibility, UI automation testing, and tooling

Tech: IBM Bluemix (Cloud), Vue.js, Angular, React, JavaScript, I18n, D3.js, Mobile Web App, Legacy Modernization

Senior Web Application Developer

Aug 2013 - Oct 2015

Preventice Solutions (now Boston Scientific) (Rochester, MN)

- Built frontend and backend systems supporting BodyGuardian, an FDA-cleared wearable cardiac monitoring device transmitting patient data to clinical care teams
- Developed the Notification Service that delivered cardiac alerts to clinical staff, a critical component in a patient-care pipeline requiring high reliability
- Shipped within FDA and HIPAA constraints, translating strict compliance requirements into reliable software delivery

Tech: Java, C#, REST APIs, ActiveMQ, Identity Management, SAML SSO, ASP.NET, SQL Server, Angular

Staff Software Engineer, Team Lead, Test Architect

May 2002 - Aug 2013

IBM Systems Group (Rochester, MN)

- Ported IBM Directory Server (LDAP) to IBM i, a commercially shipped enterprise product.
- Co-designed the test strategy and tool ecosystem for a new product line, co-leading three distributed engineering teams in Beijing to execute full product validation.
- Automated software update packaging, eliminating 500+ person-hours of manual effort annually and enabling self-service builds across teams

Tech: C++, Identity Management & Security, REST APIs, Java, JavaScript, Perl, IBM Db2, I18n

Software Engineering Intern

May 2001 - Aug 2001

Intel Corp. (Hillsboro, OR)

- Developed face-based search applications for the consumer photo market

Tech: Java, Image Processing Integration

EDUCATION

Master of Science in Computer Science | Purdue University, 2002

- Theory of Computation, Advanced Networking, Distributed Databases, OS Implementation, Cryptography

Bachelor of Science in Computer Science | Brigham Young University, 2000

- Research assistant to Dr. Tony Martinez developing neural network training model for handwritten character recognition in C